Agricultural waste management is one of the most critical environmental challenges facing India today. Every year, approximately 500 million tonnes of crop residue is generated, out of which a significant portion is burned in open fields. This practice leads to severe air pollution, soil degradation, loss of nutrients, and contributes significantly to greenhouse gas emissions. The National Capital Region (NCR) witnesses hazardous air quality levels every winter, primarily due to stubble burning in neighboring states.

Despite the environmental and health hazards, farmers often resort to burning because they lack accessible, affordable, and convenient alternatives for waste disposal. There is a disconnect between farmers who generate waste and the recycling industries that can convert this waste into valuable products such as biofuel, compost, animal feed, and building materials.

AgriTrace bridges this gap by providing a digital platform that connects farmers with collection agents and recycling facilities, creating a transparent and efficient agricultural waste management ecosystem. The platform leverages modern web technologies and cloud infrastructure to deliver a real-time, responsive, and scalable solution that can operate across diverse geographical regions.

1.1 Aim of the Project

The aim of AgriTrace is to develop a comprehensive web-based platform for tracking, managing, and recycling agricultural waste. The platform facilitates the connection between farmers, waste collection agents, and recycling administrators through a role-based system that ensures efficient workflow management from waste generation to recycling completion.

The specific aims include:

- Provide farmers with a digital tool to report and list agricultural waste for collection.
- Enable collection agents to discover, claim, and manage waste pickup tasks efficiently.
- Empower administrators to oversee the entire waste management pipeline with analytics and user management capabilities.
- Track the environmental impact through carbon credit calculations.
- Incentivize participation through a gamification and rewards system.
- Integrate secure payment processing for waste transactions.

1.2 Motivation

The motivation behind AgriTrace stems from several critical factors:

1. **Environmental Crisis:** Open burning of agricultural waste contributes to approximately 25% of air pollution in northern India during October–November. A digital platform can help divert waste from burning to recycling.